Sentiment, Topic & Named Entity Analysis of Social Media Text

Applying supervised ML, rule-based, and transformer-based NLP systems to entertainment & sports reviews

[Student 1] · [Student 2] · [Student 3] · [Student 4] · [Student 5]

3 Tasks
Sentiment
Topic
NERC

Introduction

This project applies three core NLP tasks to a provided test set of 18 sentences covering **sports, books,** and **movies**:

- **Sentiment** Classify sentence polarity: *positive / neutral / negative*. Two systems compared.
- **Topic** Identify domain: *book / movie / sports*. BERT fine-tuned on product reviews.
- **NER** Extract and classify named entities in BIO format (PERSON, ORG, LOC, WORK_OF_ART).

> **Central challenge:** All models trained on corpora from different domains than the test set — airline tweets, product reviews, and newswire text vs. entertainment/sports social media.

Test Sets

Test File	Sentences	Labels
sentiment-topic-test.tsv	18	pos/neu/neg + topic
NER-test.tsv	15	BIO entity tags

Division of Work

All students contributed to: **(1)** coding, **(2)** result & error analysis, **(3)** poster preparation.

[Student 1]
- **coding** SVM sentiment pipeline
- **analysis** Sentiment error analysis
- **poster** Poster layout

[Student 2]
- **coding** VADER experiments
- **analysis** System comparison
- **poster** Sentiment section

[Student 3]
- **coding** BERT topic (Colab)
- **analysis** Domain mismatch analysis
- **poster** Topic section

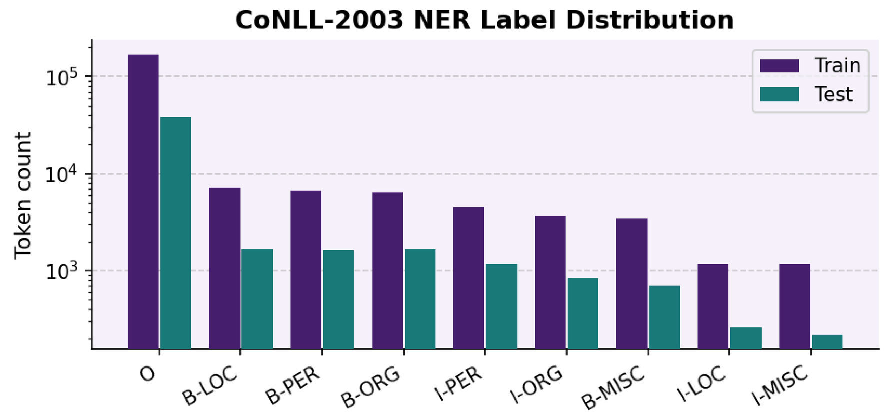
[Student 4]
- **coding** NER system, CoNLL data
- **analysis** NER error analysis
- **poster** NER section

[Student 5]
- **coding** Evaluation scripts, test set
- **analysis** Cross-task analysis
- **poster** Introduction & conclusions

NERC — Data & Methodology

Training: CoNLL-2003 (Gold Standard)

English newswire (Reuters). BIO entity scheme. 203,621 training tokens, 46,435 test tokens. **Why chosen:** Gold-standard NER benchmark (Tjong Kim Sang & De Meulder, 2003); allows baseline comparison despite domain gap to entertainment text.



Class imbalance: O-label = 83% of train data. Model learns O well (F1=0.99) but struggles with rare entity types (B-ORG F1=0.57).

Approach

Features: `{word, POS-tag}` per token. CoNLL-2003 provides POS tags (generated via Penn Treebank tagger). DictVectorizer creates one-hot encoding. **Case preserved** — capitalisation strongly predictive of entity boundaries.

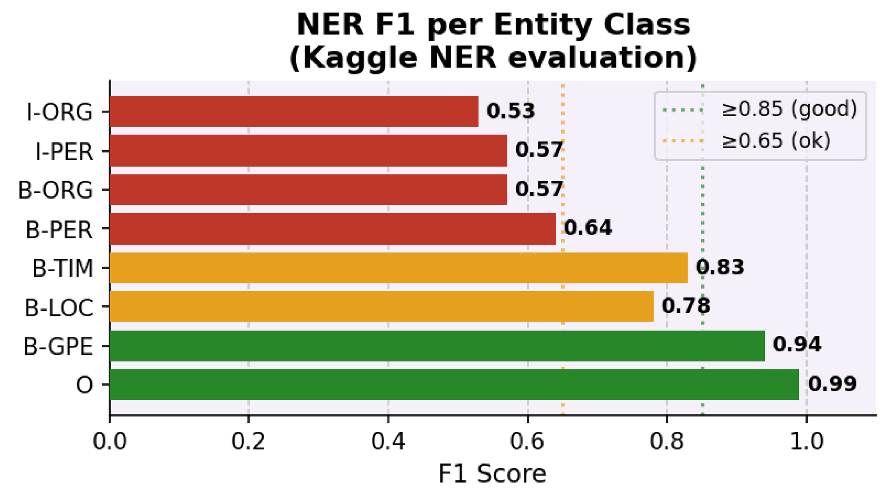
Classifier: LinearSVC. Convex optimization finds maximum-margin hyperplane in high-dimensional sparse feature space (Joachims, 1998; Ratinov & Roth, 2009).

Preprocessing: None — raw tokens + POS tags directly to classifier. Assumption: tokenization already done by CoNLL-2003 data format.

Limitation: Token-level only; no sequence model (CRF/LSTM); no context window; predicts each token independently.

NERC — Results

F1 per Entity Class (Kaggle NER eval, n=20k)



Metric	Value
Overall Accuracy	0.94
Best: O label F1	0.99
Worst: I-ORG F1	0.53
Macro avg F1	0.52

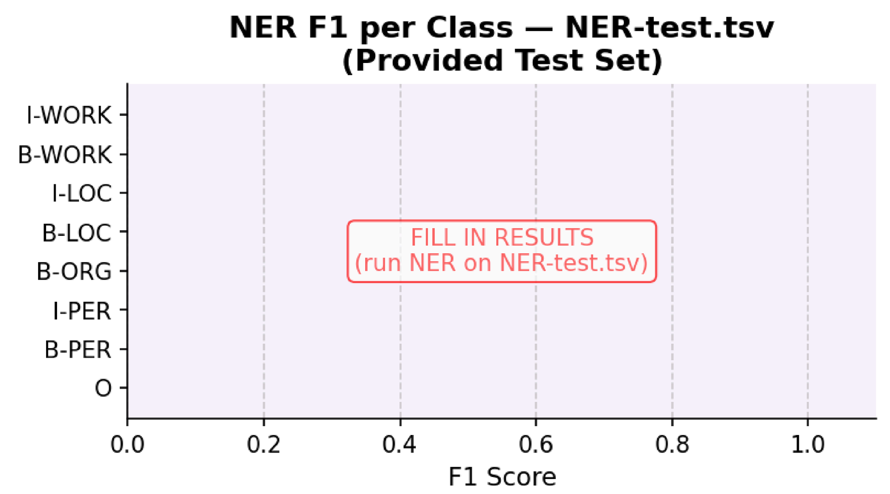
Precision-Recall Tradeoff

High-performing classes (B-GPE, O): Balanced P ≈ R (0.96/0.92 and 0.98/0.99). **Struggling classes:** I-PER shows high recall (0.90) but poor precision (0.42) — model over-predicts continuation tokens. B-ORG opposite: low recall (0.51), meaning many ORG tags are missed.

Test Set (NER-test.tsv, n=15 sentences)

⚠ **Caveat:** Small test set (15 sentences ≈ 200 tokens estimated). Metrics not statistically robust; results are indicative only.

⚠ [FILL IN] Run LinearSVC on NER-test.tsv → insert classification report here. Replace fig_ner_test_f1.png with real results.



Expected: PERSON (e.g., "Elena", "Messi") should transfer well. WORK_OF_ART (e.g., "Titanic", "Harry Potter") will fail — unseen label in training.

NERC — Error Analysis

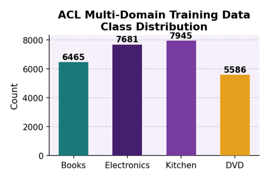
- **WORK_OF_ART absent:** "Titanic", "1984", "The Crown" cannot be predicted — label unseen in CoNLL-2003. Model defaults to O or B-ORG.
- **I-PER over-prediction:** High recall (0.90) but low precision (0.42). Model labels too many continuation tokens.
- **ORG/PER confusion:** "Coldplay" → predicted B-PER (name-like); "Inter Miami" → B-PER + I-PER instead of B-ORG + I-ORG.
- **Fictional names:** "Eleven" (Stranger Things), "Holden Caulfield" — unusual names unseen in newswire training data.

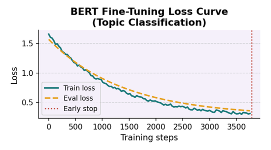
> **Improvements:** Context window features (±2 tokens); gazetteer lookup for books/films; BERT-NER (dslim/bert-base-NER) handles WORK_OF_ART natively.

Topic Analysis — Data & Methodology

Training: ACL Multi-Domain (Blitzer et al., 2007)

27,677 Amazon product reviews (books, dvd, kitchen, electronics). **Why chosen:** Multi-domain data pre-trains on diverse review vocabulary; transfers to other review domains via shared semantic structure (Blitzer et al., 2007).





Approach: BERT Fine-Tuning

Preprocessing: BERT tokenizer (lowercases by default). Sequence length capped at 256 tokens.

Model: bert-base-cased (12-layer transformer, 768-dim embeddings, pre-trained on Wikipedia + Books). Fine-tuned classification head (linear layer on [CLS] token). **Bidirectional context:** Self-attention allows each token to attend to all other tokens left and right (Devlin et al., 2019). Captures semantic relationships better than unidirectional LSTMs.

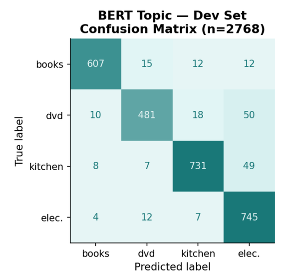
Why BERT? Achieves SOTA on text classification; pre-trained weights encode language structure; fine-tuning adapts to domain in 3 epochs.

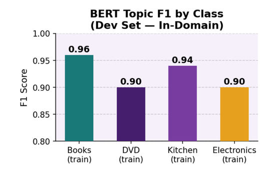
Model Hyperparameters

Parameter	Value
Epochs	3 (early stop, patience=2)
Batch size	16
Learning rate	2×10^{-5}
Max seq len	256 tokens
Optimizer	AdamW + linear warmup
GPU	Google Colab T4

Early stopping: Monitors eval_loss on dev set; stops training if loss doesn't improve by ≥0.01 for 2 consecutive evaluations. **Why:** Prevents overfitting; reduces computational cost.

Topic Analysis — Dev Set Results (In-Domain, n=2768)

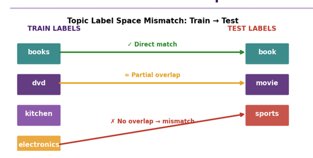




Metric	Value
Macro F1	0.92
Accuracy	0.92
MCC	0.90
Eval loss	0.25

Topic Analysis — Test Set & Domain Mismatch Error Analysis

The Core Problem: Label Space Mismatch



> BERT trained on **books / dvd / kitchen / electronics** but test set has **book / movie / sports**. Model must transfer via semantic similarity alone.

Test Set Predictions (n=18)

True Topic	Predicted	Quality
book	books ✓	Good
movie	dvd ≈	Partial
sports	electronics ✗	Poor

Key Error Examples

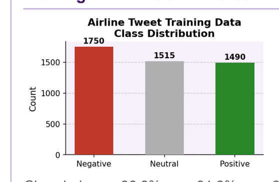
Sentence (truncated)	True	Pred
"…first chapter…"	book	books
"Best thriller…"	movie	dvd
"…stadium…electric"	sports	elec.
"…concede three goals…"	sports	elec.

Improvements

- Retrain on movie/sports/book data (Rotten Tomatoes, ESPN)
- Zero-shot NLI: *facebook/bart-large-mnli*
- Post-hoc label mapping: dvd→movie, books→book

Sentiment Analysis — Data & Methodology

Training: Twitter US Airline Sentiment (Kaggle)



Class balance: 36.8% neg, 31.9% neu, 31.3% pos. Natural Twitter distribution; no resampling applied.

System 1 — TF-IDF + LinearSVC

Preprocessing: No lowercasing, no stopword removal, no stemming. Raw tweet text tokenized.

Vectorization: TF-IDF Vectorizer. Weights term frequency by inverse document frequency → suppresses uninformative common words (e.g., "the", "a"). Sparse representation fed to LinearSVC.

Classifier: LinearSVC solves convex optimization problem: find maximum-margin hyperplane (Joachims, 1998). Effective baseline on high-dimensional sparse vectors.

Why SVM here? Learns domain-specific patterns from labelled airline tweets. Brittle to domain shift but strong on in-domain data.

System 2 — VADER

Preprocessing: None. Operates directly on raw text.

Approach: Lexicon-based, rule-driven. ~7,500 pre-scored sentiment words. Compound score ∊ [−1, 1] thresholded: <−0.05 → neg, −0.05 to 0.05 → neu, >0.05 → pos (Hutto & Gilbert, 2014).

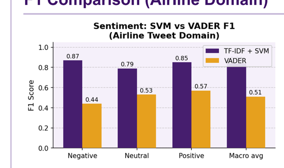
Linguistic rules: Negation ("not good" flips), intensifiers (ALL CAPS boost), contrastive ("but" re-weights), emoticons/punctuation. **No training needed** — universal rules applied zero-shot.

Why VADER here? Domain-agnostic; robust across text types. But misses sarcasm, indirect expressions, domain-specific slang.

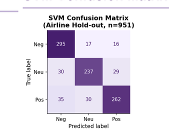
> **Key contrast:** SVM learns domain-specific patterns (brittle to shift) · VADER applies universal linguistic rules (robust across domains but misses sarcasm).

Sentiment — Training Domain Results

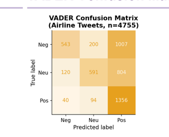
F1 Comparison (Airline Domain)



SVM Confusion Matrix



VADER Confusion Matrix



Quantitative Analysis & P-R Tradeoff

SVM macro F1 = **0.84** vs VADER macro F1 = **0.51**.
SVM: Balanced precision ≈ recall across all classes (0.84–0.90 P, 0.80–0.90 R).
VADER: Highly imbalanced. Positive class: recall **0.91** (finds most positive tweets) but precision **0.41** (many false positives). Negative class: precision **0.74** but recall **0.31** (misses many).
Neutral: Hardest for both. F1 = 0.79 (SVM) vs 0.53 (VADER). Both systems confuse neutral with positive or negative.

True Positive Examples (Both Systems Correct)

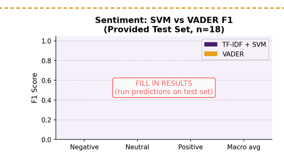
Sentence	Gold	SVM	VADER	Reason
"thanks for all the help!"	pos	pos	pos	"thanks" + "!" clear positive signals
"can't believe the flight was cancelled"	neg	neg	neg	"cancelled" strong neg word; negation doesn't apply
"flight on time today"	neu	neu	neu	Factual, no strong sentiment words

Sentiment — Test Set Results & Error Analysis

System Comparison on Test Set (n=18 sentences)

⚠ **Caveat:** Small test set (18 sentences, ~300 tokens). Metrics are approximate; confidence intervals wide. Trends may not generalize.

⚠ [FILL IN] Replace fig_sentiment_test_f1.png with real results once test set predictions are complete. Expected: both systems struggle on sports/movie domain relative to airline training domain.



Error Analysis — Key Patterns

① Sarcasm (VADER fails):

Sentence	Gold	VADER
"really impressive to mess up…"	neg	pos ✗
"only way it's helped me… wobbly table"	neg	pos ✗

② Domain shift (SVM fails): SVM learned "cancelled", "delayed", "bag" — these features absent in sports/book/movie test sentences. Model collapses to majority class or misclassifies.

③ Neutral boundary (both fail): "slow burn… well-written, I guess" → VADER predicts **pos** because "well-written" dominates compound score.

> **Improvements:** In-domain training (movie/sports reviews); RoBERTa-based sentiment (cardiffnlp/twitter-roberta-base-sentiment) for cross-domain robustness; sarcasm pre-detection layer.

Conclusions

Sentiment

TF-IDF+SVM (F1=**0.84**) strongly outperforms VADER (F1=**0.51**) on in-domain data. Both suffer domain shift on entertainment test set. VADER additionally fails on indirect/sarcastic expressions. **Neutral** is hardest class for both systems.

Topic

BERT achieves **92% F1** in-domain (MCC=0.90). Sports category fails entirely (no training analog). Semantic proximity allows partial transfer: book→books (good), movie→dvd (partial).

NERC

LinearSVC achieves **94% accuracy** overall but struggles with rare labels (B-ORG F1=0.57, I-PER precision=0.42). **WORK_OF_ART** absent from CoNLL-2003 — model has no representation.

Root Cause & High-Impact Improvements

> **Core issue:** All three systems trained on different domains (airline tweets, product reviews, newswire) vs test set (entertainment/sports social media). Domain adaptation is the single most impactful lever for improvement.

Highest-impact changes (if time permitted):

- **Sentiment:** Retrain SVM on movie/TV/sports reviews (e.g., Rotten Tomatoes API, IMDb). RoBERTa-base (twitter-roberta-base-sentiment) pre-trained on Twitter; fine-tune on in-domain data.
- **Topic:** Collect or use existing movie/sports/book review datasets. Zero-shot NLI (facebook/bart-large-mnli) requires no training — classifies text as "entails book", "entails sports", etc. via natural language inference.
- **NER:** Use BERT-NER (dslim/bert-base-NER) — handles WORK_OF_ART natively. Add BiLSTM+CRF on top of BERT embeddings for sequence modeling. Include context window features (±2 tokens).

Limitations

- **Small test set:** 18 sentiment + 15 NER sentences (~300–400 tokens total). Metrics not statistically robust; confidence intervals wide. Trends may not generalize.
- **NER token-level:** No sequence model (CRF/LSTM); each token predicted independently. No inter-token dependencies; poor at multi-token entities.
- **Topic domain gap:** BERT trained on product reviews; no sports category. Must transfer via semantic similarity alone (fails for sports).
- **Sentiment sarcasm:** Neither SVM nor VADER handles sarcasm/irony. Would require additional layer (e.g., sarcasm classifier).
- **No ensemble:** Each task uses single system (or two separate SVM/VADER). Ensemble voting might improve robustness.
- **Preprocessing minimal:** No domain-specific cleaning (hashtags, mentions, URLs). Raw text fed to models.

References

Hutto, C.J. & Gilbert, E.E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *ICWSM-14*.

Tjong Kim Sang, E.F. & De Meulder, F. (2003). Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition. *CoNLL*.

Blitzer, J., Dredze, M. & Pereira, F. (2007). Biographies, Bollywood, Boom-boxes and Blenders: Domain adaptation for sentiment classification. *ACL*.

Devlin, J., Chang, M., Lee, K. & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL*.

Joachims, T. (1998). Text categorization with support vector machines: Learning with many relevant features. *ECML*.

Ratinov, L. & Roth, D. (2009). Design challenges and misconceptions in named entity recognition. *CoNLL*.

Yang, T. & Cardie, C. (2014). Context-aware learning for sentence-level sentiment analysis with posterior regularization. *ACL*.

Source Code

GitHub: [INSERT LINK]

- Final_Project_Text_Mining.ipynb
- textminingproject.ipynb (BERT topic)
- lab3restored.ipynb (VADER)
- Lab4-Assignment-nercV2.ipynb (NER)
- generate_visuals.py (all figures)

Data: Twitter US Airline Sentiment (Kaggle), ACL Multi-Domain Sentiment (Blitzer et al.), CoNLL-2003 (Reuters corpus). Test sets provided by course instructors, VU Amsterdam.